

Developmental Observation of Teaching



Faculty Name:		Mukto Akash	
Observer Name:		David MacLennan	
Chair:		Marilyn Powers	
Program Name:		MATH73235, Advanced Topics in Mathematics, Section 1	
Date of Observation (mm/dd/yyyy):		06/17/2025	
Course (Code, Name, and Section):		MATH73235, Advanced Topics in Mathematics, Section 1	
Campus, Room Number:		In-Person	
Course Outcome(s) Addressed in this Lesson:		2.12 Use Laplacian in Polar Coordinates to solve a circular membrane problem [GA: PA2 Methodology]	
Week of Semester:		7	
List of materials sent in advance:		None	
Type of classroom setting (e.g. Zoom, Teams, in person, etc.):		In-Person	
Attendance taken by faculty: (Yes, Not Observed):	Yes	Attendance Method Used:	Visual. You noted who was missing.
Students enrolled in the class: 10	Students at start of class: 8	Students who arrived late: 0	Students at end of observation: 8

Instructions for Faculty

Dear Faculty,

1. Please read through your observation and reflect on the comments and suggestions.
2. After you are finished reading through, open the document that was emailed to you by your observer called "Reflections on My Teaching Observation" and save the file adding your name and the course observed.
3. Add one comment, question, and commitment to each section of the table as per the example. These reflections will be the start of your conversation when you meet your observer for the debrief.
4. Email the reflections back to your observer within 3 days.
5. Your observer will have arranged a meeting with you to discuss the observation and share further insights.
6. After the meeting occurs, the observation will be sent to your Chair with you cc'd.
7. Our goal is to have the observation to the Chair within two weeks of your first contact with your observer.

If you have any questions about this process please email : teachingandlearning@conestogac.on.ca

Summary of the Observation

Strategies that Supported Student Learning

Knowledgeable - It was abundantly clear that you knew the subject matter thoroughly. Not only did you explain as you were writing, you also pointed out some interesting aspects to some of the formulae.

Relationships with Students - It was clear that you have the respect of the students. You used student names and also engaged in some light humour before getting into today's topic. Students were observed to be engaged intently with note-making or solving as you spoke.

Demonstration and Time for Tasks - You paused at appropriate times to allow for reflection, note-making and the occasional formula to solve. You also gave students time to respond to your questions.

Opportunities to Enhance Practice as a Professional Educator

Use the BOPPPS Lesson Planning Template - The BOPPPS planning model actually prompts you to include a bridge in activity and opportunities for student engagement/ interaction. Here are two links to more information:

<https://tlconestoga.ca/creating-a-bopppps-lesson-plan/>

<https://tlconestoga.ca/content-based-bridge-ins/>

Use of Guided Video or Video segments - You may wish to consider using a video or video segment to introduce or review a topic. Before showing the video or video segment, give students some "look-fors" to get them to focus on the important points you want them to see. I found this video on Polar Coordinates as an example:

https://youtu.be/aSdaT62ndYE?si=pdI_yhH1-1NBMGMe

Here is a link to more information in using a video in class:

<https://tlconestoga.ca/lights-camera-learn-best-practices-for-sharing-videos-during-class/>

Student Engagement Ideas - You made use of Slido which has an open response section. Mentimeter also has this feature; you may wish to try it as free versions of platforms usually start putting restrictions on to encourage purchasing. Also, pausing for "key take-aways" in pairs can help with consolidation and potential questions. Here are links to Mentimeter and Think-Pair-Share:

<https://tlconestoga.ca/mentimeter/>

<https://tlconestoga.ca/think-pair-share-a-quick-win-teaching-tip/>

Section A - Elements of Lesson/Delivery

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
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Legend:

Y = Yes, SW = Somewhat, N = No, NA = Non Applicable

Set Up

QA1. Class started on time*	Y	You mentioned that you normally start ten minutes late. Always good to start on time.
QA2. Materials, room, and technology were ready	Y	You had Slido on screen for anonymous comments and questions. This was then turned off on screen so the white-board could be used. You checked Slido on your computer for any questions or comments during class.
QA3. Students welcomed	Y	
QA4. Agenda on board and explained	Y	You wrote a brief agenda on the white-board.

Bridge-in

QA5. Gained attention and energy	N	A bridge-in activity was not used. You may wish to consider a brief guided video segment or Mentimeter Quiz. See the summary comments for more ideas.
QA6. Established subject-specific relevance	NA	

Outcomes (expected learning)

QA7. Tied lesson to course outcomes,* Instructional Plan, course evaluations, and how to be successful*	Y	The class related directly to Learning Outcome 2.12 (Polar Coordinates).
QA8. Tied lesson to Essential Employability Skills and program outcomes (if applicable)	NA	
QA9. Alerted students to any sensitive subject matter	NA	

Section A - Elements of Lesson/Delivery

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Pre-Assessment

QA10. Activated and determined student level of prior learning (e.g. knowledge, skills, attitudes)	N	See the summary comments regarding the BOPPPS lesson planning template.
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Presentation of Content for Learning

QA11. Organized*, well-paced*, and well-planned*	Y	You were clearly ready for class. You may wish to work in brief pauses for paired interaction, video, or other strategies that provide a different teaching modality.
QA12. Explained content effectively*	Y	You talked through the formulae as you wrote on the white-board.

Participatory Activities

QA13. Co-constructed knowledge	NA	Not part of time observed
QA14. Applied content	NA	
QA15. Provided variety*	NA	

Opportunities to Practice

QA16. Controlled	NA	Not part of time observed
QA17. Semi-Controlled	NA	
QA18. Free	NA	

Post-Assessment

QA19. Confirmed level/degree of student learning	NA	Not part of time observed
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Summary

QA20. Brought deliberate closure	NA	Not part of time observed
QA21. Emphasized key learning	NA	

Section B - Clarity and Subject Matter Expertise

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
QB1. Demonstrated a thorough knowledge of course concepts*	Y	This was clearly evident.
QB2. Used field-specific terminology effectively	Y	
QB3. Related course material to real-world applications*	NA	
QB4. Discussed recent/future developments in the field*	NA	
QB5. Placed content in context as appropriate (e.g. historically, internationally, geographically, philosophically, politically)	NA	
QB6. Limited asides / remained on-topic	Y	

Section C - Delivery Skills, Enthusiasm and Rapport

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
QC1. Demonstrated interest and enthusiasm in the subject*	Y	
QC2. Presented material in an interesting way* (i.e. used a variety of learning modalities to engage learners)	SW	Most was done through white-board formulae. Consider some tasks embedded for students to try. Video can also be helpful and provide visual examples.
QC3. Motivated students to do their best work* and encouraged participation*	Y	You posed some questions to the class. Students were given time to work on solving a problem regarding inverse transformation. See the summary comments for more ideas.
QC4. Used students' names (including preferred pronouns) during class	Y	You knew the students' names. If it helps, name tents are a good idea at the beginning of the semester.
QC5. Handled unexpected events and comments calmly and empathetically	NA	
QC6. Used voice effectively (e.g. volume, tone, pace, emphasis)	Y	
QC7. Delivery including proxemics, gestures, and oculesics were supportive	Y	
QC8. Facial expressions and demeanor were conducive to student comfort	Y	Students were obviously comfortable with you.
QC9. Energy level was engaging	Y	You kept the class moving at a good pace.

Section D - Teaching Skills

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
QD1. Stimulated discussions* and invited opinions*	SW	You posed questions of the class. See the summary comments for other ways to engage students.
QD2. Generated widespread participation in all aspects of the learning opportunity	Y	Yo had students taking notes and solving equations. See the summary comments for more ideas.
QD3. Ensured pairs or groups were functional when tasks were assigned (e.g. circulated, coaching, problem-solving as appropriate)	NA	
QD4. Managed classroom behaviours in a way that appeared respectful and promoted a positive learning environment, addressing any concerns related to discrimination and harassment.	NA	No issues
QD5. Provided meaningful opportunities for quiet processing	Y	You gave time for students to reflect and try to solve equations.
QD6. Provided breaks and time for student/professor regeneration/reflection	Y	
QD7. Provided adequate instructions for any activities or tasks, and modeled targeted skills and/or tasks	Y	
QD8. Modeled targeted skills and/or tasks	Y	This was done quite well. You demonstrated the answer after students had an opportunity to solve the equations.
QD9. Provided pauses, transitions and sectioning to aid comprehension, note-taking, and attention*	Y	This was done well. You gave pauses for students to reflect, take notes and try problems.
QD10. Managed the asking of and responding to questions effectively	Y	
QD11. Explained content using language that students understood*	Y	Students were observed interacting with the terms.
QD12. Used comprehension checks to ensure class understood material*	NA	You may wish to consider a task to take up or a brief anonymous quiz.
QD13. Provided helpful feedback*	Y	

Section D - Teaching Skills

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
QD14. Textbook/reading materials helped students*	SW	You commented that the topic being covered today was not covered as well in the text and that you were doing a "deeper dive" into the concepts.
QD15. The delivery connected this course learning to the students' wider semester/program experience*	Y	
QD16. Technology was used to help students learn* including:	SW	You used Slido briefly for comments an questions. See the summary comments for more ideas.

Section E - Diversity, Inclusion, and Accessibility

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
QE1. Delivery took into account the accessibility needs of students and Universal Design for Learning (e.g. closed captioning, material design, posting materials to eConestoga, accessibility of learning materials, other _____)	SW	The main method was using formulae. You did have times for students to try tasks on their own. Consider visuals such as through related video segments.
QE2. Provided electronic copies of resources on eConestoga prior to class*	NA	Not observed

Delivery and materials were accessible and in keeping with the Accessibility for Ontarians with Disabilities Act (AODA), Ontario Human Rights (OHR) and The Freedom of Information and Protection of Privacy Act (FIPPA)

QE3. respect for individual needs	NA	None noted
QE4. board writing	Y	This was very clear.
QE5. presented a professional classroom environment (e.g. sufficient lighting, appropriate attire, visibility, etc.)	Y	
QE6. seating arrangements	Y	
QE7. organization of/options for activities	NA	
QE8. freedom of movement for physical needs	NA	No issues noted
QE9. maintenance of student privacy and confidentiality	NA	

Section F - Legal

*Represents items aligned with the Formal SAT instrument

	Observed Y/SW/N/NA	Observations and Suggestions
QF1. Demonstrated professional practices and behaviours including use of any required personal protective equipment (PPE)	Y	No issues
QF2. Interactions were in keeping with the Ontario Human Rights code (OHR)	Y	No issues
QF3. Complied with copyright and acknowledged sources used	Y	No issues

Glossary

Active Learning: Learning opportunities provided to students that allow them to process and manipulate the content so that they build knowledge, skills, and attitudes appropriate to the learning outcomes for the course. Sitting passively and listening to the professor is not active learning though it may be an important sub-component of a classroom experience.

AODA: Accessibility for Ontarians with Disabilities Act. All educators are required to provide accessible learning opportunities and materials.

[See Accessibility Services Faculty Information](#)

[See Creating Accessible Materials.](#)

Experiential Learning: The Ministry of Colleges and Universities (MCU) asks that students be exposed to authentic demands that improve their employability, interpersonal skills, and transition to the workforce. In the classroom, this can involve purposeful and meaningful activities that simulate the intended workplace.

Instructional Plan: An Instructional Plan is a detailed schedule outlining course delivery elements. It guides faculty delivery and student planning and preparation. It is based on a college-wide template that ensures that a) faculty have determined the best way to deliver a course over the time-period available, b) that evaluations have been distributed in a way that allows for marking and feedback, c) students can anticipate and prepare for the key concepts being delivered each class, and d) the date for each evaluation is clearly communicated to students and other stakeholders.

Learning-Centered Practice: This is an approach in which the faculty focusses on delivering a course where significant learning opportunities are the focus. The faculty is not the “sage on the stage” but rather “a guide on the side” helping learners develop knowledge, skills, and attitudes related to the course outcomes. Assessments are developed that require, encourage and foster further learning.

Occulesics: The study of the use of the eyes in communication. It is important that the professor make equitable eye contact with all learners and engage the farthest corners of the class with eye contact.

Ontario Human Rights (OHR) Code: The Code provides for equal rights and opportunities and freedom from discrimination.

[See the protected grounds at Ontario Human Rights Code](#)

Outcomes-Based Education: This approach identifies outcomes that the successful student will be able to achieve by the end of the course and aligns learning opportunities and assessments to these stated outcomes.

Pedagogy: The methods and practice of teaching. One engages in pedagogy when one is planful and deliberate in offering learning opportunities.

PPE: Personal Protective Equipment is required in some areas. Your Manager will inform you. PPE provides protection for heads, ears, eyes, skin, feet, and the respiratory system. PPE provides protection from hazards and falls.

Proxemics: Proxemics are the placement of the body and other items (e.g., furniture, screens) in the available space. It would be poor proxemics to stand in one corner of the class all session or to pace back and forth. It is good proxemics to be visible to the students and to change sides occasionally so as to equalize attention. It is also good proxemics to stand close to and gesture to items on the board while you speak about them.

Realia: Items from the working world that are brought into the class so students can interact with them and see them to scale and in 3-dimension. These can include items such as tools or equipment that students will later use in the work place. They can be passed around, demonstrated or used in the class.

Respect: In partnership with civility and empathy, respect is a key component of a functional learning environment. It is challenging to respect each student and the class as a community when some do show the respect we expect, yet faculty who score high on respect on the Student Appraisals of Teaching (SAT) usually score well overall. The opposite is also true.

Triggering Topics: Some students have very painful personal experiences related to rape, drugs, abortion, death, car accidents, alcohol abuse, cancer and other sensitive topics. It is important to let them know if such a theme is introduced or emerges and to never speak flippantly about such topics.

Universal Design for Learning (UDL): Universal Design for Learning is a framework to improve and optimize teaching and learning for all people based on scientific insights into how humans learn. It proposes that if you think about the needs of students with disabilities or other challenges and think about how to adapt your teaching and materials to help them that you can create a delivery that is universally more helpful. Just as the cut-outs on curbs for sidewalks was done for wheelchairs but helped people on bikes and with strollers so can your making enhancements to your teaching help everyone. See UDL Video from CAST.

[See UDL Video from CAST.](#)